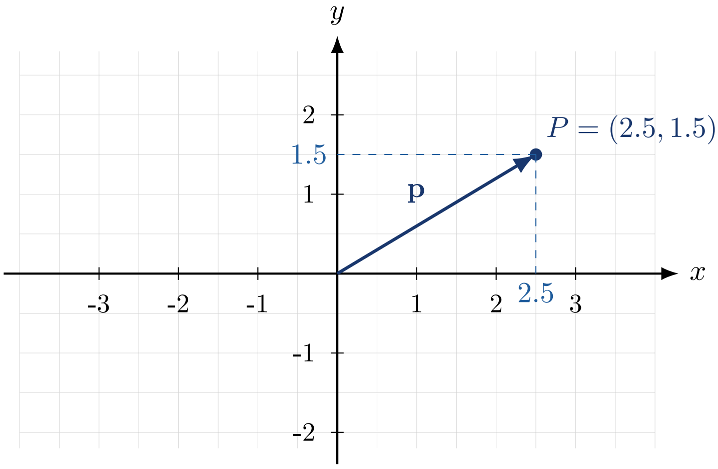
Lecture Notes on Cartesian Space

Coordinates, vectors, distance, lines, planes, and basic geometry



Prepared as introductory mathematical lecture notes

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1 Why Cartesian space matters

Cartesian space is the mathematical language that lets us represent geometry with numbers. Instead of saying that a point is "somewhere to the right and a bit above", we write something like

$$P = (3, 2).$$

This simple idea is the foundation of analytic geometry, mechanics, computer graphics, data analysis, numerical simulations, robotics, machine learning, and many parts of engineering.

Main idea

A Cartesian space is a space in which positions are described by ordered lists of numbers called coordinates. The most common examples are

$$\mathbb{R}, \qquad \mathbb{R}^2, \qquad \mathbb{R}^3, \qquad \mathbb{R}^n.$$

The notation $\mathbb{R}^n$ means: all ordered n-tuples of real numbers.

The point of Cartesian geometry is not only drawing pictures. The deeper idea is this:

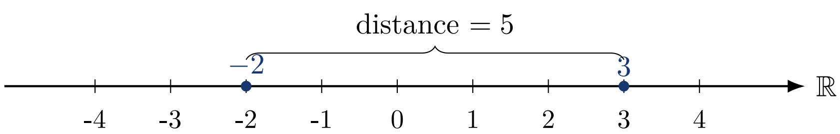
$$\text{geometry} \quad \longleftrightarrow \quad \text{algebra.}$$

A geometric question becomes an algebraic calculation. For example:

- the distance between points becomes a formula;
- a line becomes an equation;
- an angle becomes a dot product;
- a plane becomes a linear equation;
- a transformation becomes a matrix.

2 The real line and one-dimensional space

The simplest Cartesian space is the real line $\mathbb{R}$. Each real number corresponds to exactly one point on a line.



If $a, b \in \mathbb{R}$, the distance between them is

$$d(a, b) = |b - a|.$$

The absolute value is necessary because distance cannot be negative.

Example 1: Distance on a line

Find the distance between $a = -4$ and $b = 7$.

$$d(a, b) = |7 - (-4)| = |11| = 11.$$

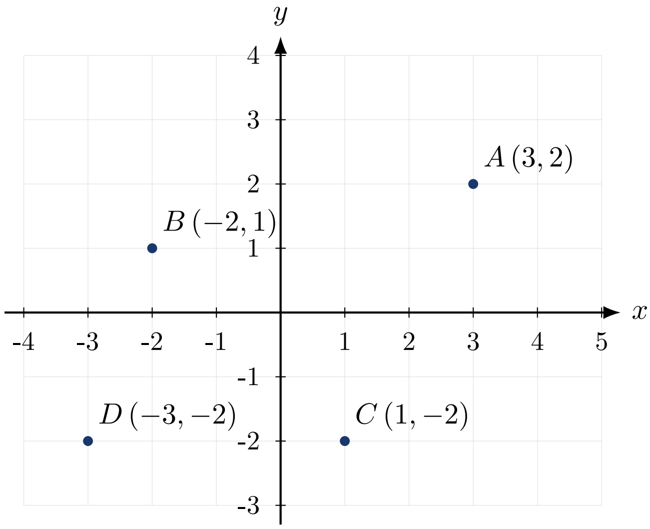
The distance is 11 units.

3 The Cartesian plane $\mathbb{R}^2$

The Cartesian plane consists of ordered pairs of real numbers:

$$\mathbb{R}^2 = \{(x, y) : x \in \mathbb{R},\ y \in \mathbb{R}\}.$$

The first coordinate is usually called the x-coordinate, and the second coordinate is called the y-coordinate.



Coordinate axes and origin

In $\mathbb{R}^2$:

- the horizontal axis is the x-axis;
- the vertical axis is the y-axis;
- the point $(0, 0)$ is the origin;
- the coordinates (x, y) tell us how far to move in each direction.

A point such as $(3, 2)$ means: start at the origin, move 3 units in the positive x-direction, then move 2 units in the positive y-direction.

Common mistake

The order of coordinates matters:

$$(3, 2) \neq (2, 3).$$

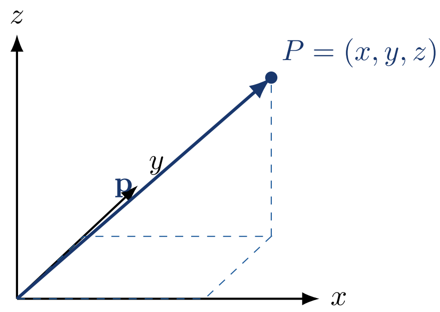
The first coordinate always refers to the first axis, and the second coordinate always refers to the second axis.

4 Three-dimensional space $\mathbb{R}^3$

Three-dimensional Cartesian space consists of ordered triples:

$$\mathbb{R}^3 = \{(x, y, z) : x, y, z \in \mathbb{R}\}.$$

The third coordinate z represents a third independent direction.



In physics, $\mathbb{R}^3$ is often used as a mathematical model of physical space. In computer graphics, it is used to represent objects, cameras, lights, and surfaces.

5 The general Cartesian space $\mathbb{R}^n$

For a positive integer n, the Cartesian space $\mathbb{R}^n$ is the set of all ordered n-tuples of real numbers:

$$\mathbb{R}^n = \{(x_1, x_2, \ldots, x_n) : x_1, x_2, \ldots, x_n \in \mathbb{R}\}.$$

The numbers $x_1, x_2, \ldots, x_n$ are called coordinates or components.

Dimension

The number n is called the dimension of $\mathbb{R}^n$. It tells us how many independent coordinates are needed to specify a point.

Examples:

$$\begin{aligned} \mathbb{R}^1 &= \mathbb{R} && \text{one coordinate,} \\ \mathbb{R}^2 &= \{(x, y)\} && \text{two coordinates,} \\ \mathbb{R}^3 &= \{(x, y, z)\} && \text{three coordinates,} \\ \mathbb{R}^{100} &= \{(x_1, \ldots, x_{100})\} && \text{one hundred coordinates.} \end{aligned}$$

High-dimensional spaces

We cannot draw $\mathbb{R}^{100}$ in the usual way, but we can still compute in it. In data analysis, a point in $\mathbb{R}^{100}$ may represent an object described by 100 numerical features.

6 Points and vectors

In elementary geometry, points and vectors are often drawn similarly, but conceptually they are different.

Point versus vector

A point describes a position. A vector describes a displacement, direction, or quantity with components.

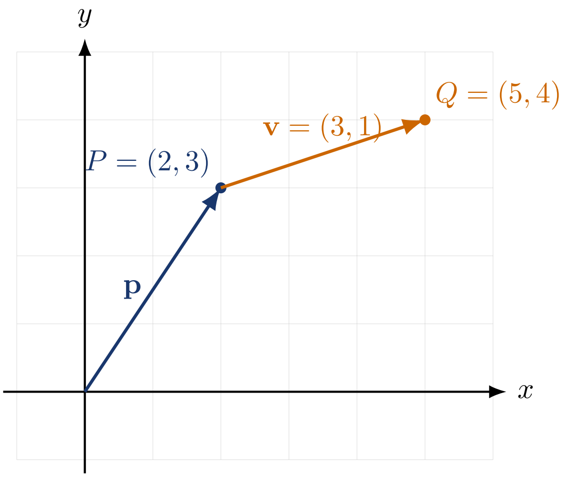
A point may be written as

$$P = (2, 3),$$

while a vector may be written as

$$\mathbf{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad \text{or} \quad \mathbf{v} = (2, 3).$$

The same coordinates may appear in both notations, but the interpretation is different.



If a vector starts at $P = (x_1, y_1)$ and ends at $Q = (x_2, y_2)$, then its components are

$$\overrightarrow{PQ} = (x_2 - x_1,\ y_2 - y_1).$$

Example 2: Vector from one point to another

Let $P = (2, -1)$ and $Q = (7, 3)$. Then

$$\overrightarrow{PQ} = (7 - 2,\ 3 - (-1)) = (5, 4).$$

So moving from P to Q means moving 5 units in the x-direction and 4 units in the y-direction.

7 Vector operations

Let

$$\mathbf{u} = (u_1, u_2, \ldots, u_n), \qquad \mathbf{v} = (v_1, v_2, \ldots, v_n).$$

Then vector addition is defined componentwise:

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n).$$

Scalar multiplication is also componentwise:

$$c\mathbf{v} = (cv_1, cv_2, \ldots, cv_n).$$

Example 3: Adding and scaling vectors

Let $\mathbf{u} = (2, -1, 4)$ and $\mathbf{v} = (3, 5, -2)$. Then

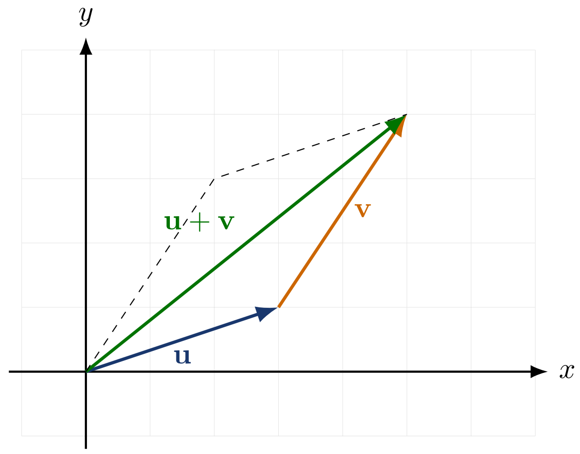
$$\mathbf{u} + \mathbf{v} = (5, 4, 2),$$

and

$$2\mathbf{u} = (4, -2, 8).$$

Geometrically:

- $\mathbf{u} + \mathbf{v}$ means doing one displacement after another;
- $c\mathbf{v}$ stretches, shrinks, or reverses the vector depending on c.



8 Standard basis vectors

In $\mathbb{R}^2$, the standard basis vectors are

$$\mathbf{e}_1 = (1, 0), \qquad \mathbf{e}_2 = (0, 1).$$

In $\mathbb{R}^3$, they are

$$\mathbf{e}_1 = (1, 0, 0), \qquad \mathbf{e}_2 = (0, 1, 0), \qquad \mathbf{e}_3 = (0, 0, 1).$$

In $\mathbb{R}^n$, the vector $\mathbf{e}_i$ has a 1 in position i and zeros elsewhere.

Every vector in $\mathbb{R}^n$ can be written as a linear combination of the standard basis vectors:

$$(x_1, x_2, \ldots, x_n) = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n.$$

Example 4: Basis decomposition

In $\mathbb{R}^3$,

$$(4, -2, 7) = 4(1, 0, 0) - 2(0, 1, 0) + 7(0, 0, 1).$$

That is,

$$(4, -2, 7) = 4\mathbf{e}_1 - 2\mathbf{e}_2 + 7\mathbf{e}_3.$$

9 Length, norm, and distance

The length of a vector $\mathbf{v} = (v_1, \ldots, v_n)$ is called its norm and is defined by

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}.$$

This formula is the Pythagorean theorem generalized to n dimensions.

Euclidean distance

The distance between two points

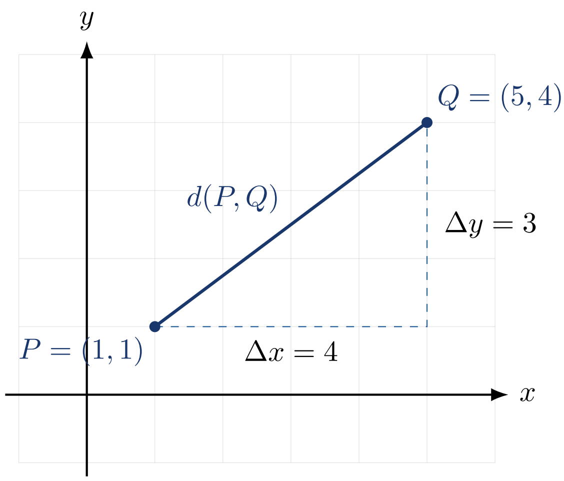
$$P = (p_1, \dots, p_n), \qquad Q = (q_1, \dots, q_n)$$

is

$$d(P, Q) = \sqrt{(q_1 - p_1)^2 + \cdots + (q_n - p_n)^2}.$$

Equivalently,

$$d(P, Q) = \|Q - P\|.$$



Example 5: Distance in the plane

Let $P = (1, 1)$ and $Q = (5, 4)$. Then

$$d(P, Q) = \sqrt{(5-1)^2 + (4-1)^2} = \sqrt{4^2 + 3^2} = 5.$$

10 Midpoints and weighted averages

The midpoint of two points $P, Q \in \mathbb{R}^n$ is

$$M = \frac{P + Q}{2}.$$

If

$$P = (p_1, \dots, p_n), \qquad Q = (q_1, \dots, q_n),$$

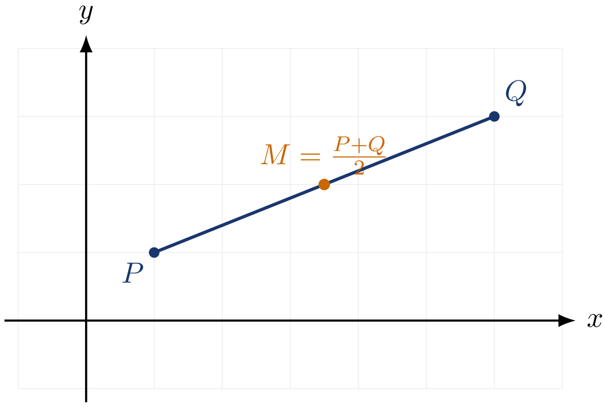
then

$$M = \left(\frac{p_1 + q_1}{2}, \dots, \frac{p_n + q_n}{2} \right).$$

More generally, for $0 \le t \le 1$, the point

$$(1 - t)P + tQ$$

lies on the segment from P to Q.



Example 6: Midpoint

For $P = (-2, 5)$ and $Q = (4, -1)$,

$$M = \left(\frac{-2+4}{2}, \frac{5+(-1)}{2} \right) = (1, 2).$$

11 Dot product and angle

For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, the dot product is

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n.$$

The dot product connects algebra with geometry:

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \, \|\mathbf{v}\| \cos\theta,$$

where θ is the angle between the two vectors.

Orthogonality

Two nonzero vectors are perpendicular, or orthogonal, if

$$\mathbf{u} \cdot \mathbf{v} = 0.$$

Example 7: Checking perpendicularity

Let $\mathbf{u} = (2, 3)$ and $\mathbf{v} = (3, -2)$. Then

$$\mathbf{u} \cdot \mathbf{v} = 2 \cdot 3 + 3 \cdot (-2) = 6 - 6 = 0.$$

Therefore, the vectors are perpendicular.

The angle between two nonzero vectors can be found from

$$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \, \|\mathbf{v}\|}.$$

12 Lines in Cartesian space

A line can be described in several equivalent ways.

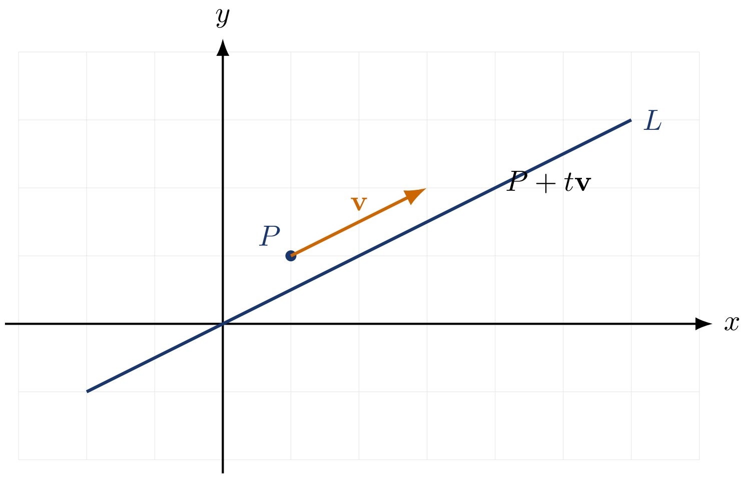
12.1 Line in parametric form

If $P \in \mathbb{R}^n$ is a point on the line and $\mathbf{v} \in \mathbb{R}^n$ is a nonzero direction vector, then the line is

$$L = \{P + t\mathbf{v} : t \in \mathbb{R}\}.$$

This means that every point on the line has the form

$$\mathbf{x} = P + t\mathbf{v}.$$



Example 8: Parametric line

A line through $P = (1, 2)$ with direction $\mathbf{v} = (3, -1)$ is

$$(x, y) = (1, 2) + t(3, -1), \qquad t \in \mathbb{R}.$$

Equivalently,

$$x = 1 + 3t, \qquad y = 2 - t.$$

12.2 Line in the plane as an equation

In $\mathbb{R}^2$, a line can often be written as

$$ax + by = c.$$

The vector

$$\mathbf{n} = (a, b)$$

is perpendicular to the line. It is called a normal vector.

Example 9: Normal vector

The line

$$2x - 3y = 6$$

has normal vector

$$\mathbf{n} = (2, -3).$$

Any direction vector of the line must be perpendicular to $\mathbf{n}$. One possible direction vector is

$$\mathbf{v} = (3, 2),$$

because

$$(2, -3) \cdot (3, 2) = 6 - 6 = 0.$$

13 Planes in $\mathbb{R}^3$

A plane in $\mathbb{R}^3$ can be described by one point and two independent direction vectors:

$$\mathbf{x} = P + s\mathbf{u} + t\mathbf{v}, \qquad s, t \in \mathbb{R}.$$

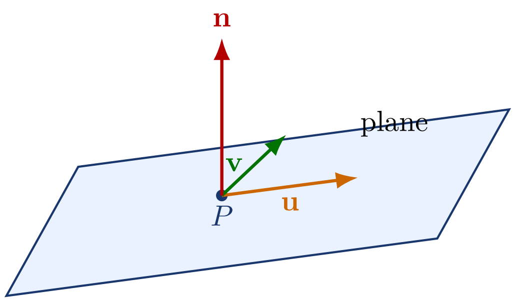
It can also be described by a linear equation:

$$ax + by + cz = d.$$

The vector

$$\mathbf{n} = (a, b, c)$$

is perpendicular to the plane.



Example 10: Plane equation

The plane

$$3x - y + 2z = 5$$

has normal vector

$$\mathbf{n} = (3, -1, 2).$$

A point lies on this plane exactly when its coordinates satisfy the equation. For example, $P = (1, 0, 1)$ lies on it because

$$3 \cdot 1 - 0 + 2 \cdot 1 = 5.$$

14 Circles, spheres, and simple surfaces

Cartesian coordinates are especially useful for describing geometric shapes by equations.

14.1 Circle in the plane

A circle with center $C = (a, b)$ and radius $r > 0$ is

$$(x - a)^2 + (y - b)^2 = r^2.$$

This equation says: the distance from (x, y) to (a, b) is r.

Example 11: Circle

The circle with center $(2, -1)$ and radius 3 is

$$(x - 2)^2 + (y + 1)^2 = 9.$$

14.2 Sphere in three-dimensional space

A sphere with center $C = (a, b, c)$ and radius r is

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = r^2.$$

Geometric interpretation

Equations like

$$x^2 + y^2 = 1$$

do not merely define algebraic sets. They describe geometric objects: here, all points at distance 1 from the origin in $\mathbb{R}^2$.

15 Coordinate transformations

A transformation maps points or vectors to new points or vectors. Some of the most important transformations are translations, scalings, rotations, and linear transformations.

15.1 Translation

A translation adds the same vector to every point:

$$T(P) = P + \mathbf{a}.$$

For example, if $\mathbf{a} = (3, -1)$, then

$$T(x, y) = (x + 3, y - 1).$$

Translation changes position but preserves distances and angles.

15.2 Scaling

Uniform scaling by a factor c maps

$$\mathbf{x} \mapsto c\mathbf{x}.$$

If $c > 1$, objects get larger; if $0 < c < 1$, objects get smaller; if $c < 0$, there is also a reversal through the origin.

15.3 Rotation in the plane

A counterclockwise rotation by angle θ about the origin is represented by

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

For example, rotation by $90°$ maps

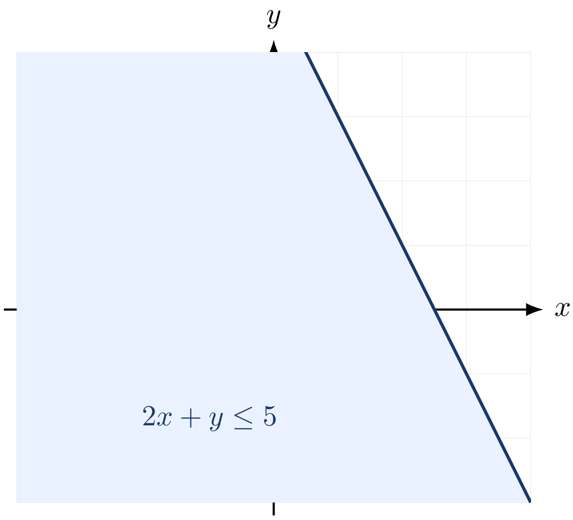
$$(x, y) \mapsto (-y, x).$$

16 Subsets, regions, and inequalities

Not every set in Cartesian space is the whole space. Very often we study subsets.

Examples in $\mathbb{R}^2$:

$$\begin{aligned} &\{(x,y) : y = x^2\} && \text{a parabola,} \\ &\{(x,y) : x^2 + y^2 \le 1\} && \text{a closed disk,} \\ &\{(x,y) : x > 0,\ y > 0\} && \text{the first quadrant,} \\ &\{(x,y) : 2x + y \le 5\} && \text{a half-plane.} \end{aligned}$$



An inequality often describes a whole region rather than a curve or surface.

17 Why Cartesian space is useful in applications

17.1 Physics

A particle position in space is often written as

$$\mathbf{r}(t) = (x(t), y(t), z(t)).$$

Velocity and acceleration are then derivatives of this vector function:

$$\mathbf{v}(t) = \mathbf{r}'(t), \qquad \mathbf{a}(t) = \mathbf{r}''(t).$$

Cartesian coordinates make motion calculable.

17.2 Computer graphics

A vertex of a triangle may be represented as a point in $\mathbb{R}^3$:

$$P = (x, y, z).$$

Transformations such as translation, rotation, and scaling are applied to many vertices to move or deform objects.

17.3 Data science

A data object can be represented as a vector. For example, a house might be encoded by

$$\mathbf{x} = (\text{area}, \text{rooms}, \text{age}, \text{distance to center}).$$

Then the house is a point in $\mathbb{R}^4$. Similarity between houses can be measured by distance or angle.

17.4 Optimization

Many optimization problems ask us to choose a point

$$\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$$

that minimizes or maximizes some function subject to constraints. Cartesian space is the natural setting for such problems.

18 Worked examples

Example 12: From points to vector and distance

Let $A = (-1, 2)$ and $B = (5, -1)$.

(a) Find $\overrightarrow{AB}$.

(b) Find $d(A, B)$.

Solution:

$$\overrightarrow{AB} = B - A = (5 - (-1), -1 - 2) = (6, -3).$$

Therefore,

$$d(A, B) = \left\|\overrightarrow{AB}\right\| = \sqrt{6^2 + (-3)^2} = \sqrt{45} = 3\sqrt{5}.$$

Example 13: Equation of a line

Find a parametric equation of the line through $P = (2, 1)$ and $Q = (5, 7)$.
A direction vector is

$$\mathbf{v} = Q - P = (3, 6).$$

Thus the line is

$$(x, y) = (2, 1) + t(3, 6), \qquad t \in \mathbb{R}.$$

Equivalently,

$$x = 2 + 3t, \qquad y = 1 + 6t.$$

Example 14: Orthogonal vectors

Find a vector perpendicular to $\mathbf{u} = (4, -1)$.
One simple method in $\mathbb{R}^2$ is to swap coordinates and change one sign:

$$\mathbf{v} = (1, 4).$$

Indeed,

$$(4, -1) \cdot (1, 4) = 4 - 4 = 0.$$

So $\mathbf{v} = (1, 4)$ is perpendicular to $\mathbf{u}$.

Example 15: Plane membership

Check whether $P = (2, -1, 3)$ lies on the plane

$$x + 2y - z = -3.$$

Substitute the coordinates:

$$2 + 2(-1) - 3 = 2 - 2 - 3 = -3.$$

The equation is satisfied, so P lies on the plane.

19 Common mistakes and how to avoid them

Mistake 1: Confusing coordinates with distance

The point $(3, 4)$ is not "the distance 3 and 4". Its distance from the origin is

$$\sqrt{3^2 + 4^2} = 5.$$

Coordinates describe components; distance is computed from components.

Mistake 2: Forgetting the order of coordinates

In $\mathbb{R}^3$, the points

$$(1, 2, 3), \quad (1, 3, 2), \quad (2, 1, 3)$$

are generally different points. Coordinate order is part of the information.

Mistake 3: Treating a point and a vector as always identical

The coordinates may look the same, but the meaning may differ. A point is a location; a vector is a displacement or quantity.

Mistake 4: Ignoring dimension

You cannot add vectors from different spaces:

$$(1, 2) + (3, 4, 5)$$

is not defined, because the dimensions do not match.

20 Summary of key formulas

Concept	Formula
Point in $\mathbb{R}^n$	$(x_1, x_2, \ldots, x_n)$
Vector from P to Q	$\overrightarrow{PQ} = Q - P$
Vector addition	$\mathbf{u} + \mathbf{v} = (u_1 + v_1, \ldots, u_n + v_n)$
Scalar multiplication	$c\mathbf{v} = (cv_1, \ldots, cv_n)$
Norm	$\lVert\mathbf{v}\rVert = \sqrt{v_1^2 + \cdots + v_n^2}$
Distance	$d(P, Q) = \lVert Q - P\rVert$
Midpoint	$M = (P + Q)/2$
Dot product	$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + \cdots + u_n v_n$
Angle	$\cos\theta = \dfrac{\mathbf{u} \cdot \mathbf{v}}{\lVert\mathbf{u}\rVert \, \lVert\mathbf{v}\rVert}$
Parametric line	$\mathbf{x} = P + t\mathbf{v}$
Plane in $\mathbb{R}^3$	$ax + by + cz = d$
Circle in $\mathbb{R}^2$	$(x - a)^2 + (y - b)^2 = r^2$
Sphere in $\mathbb{R}^3$	$(x - a)^2 + (y - b)^2 + (z - c)^2 = r^2$

21 Exercises

Basic exercises

1. Plot mentally the points $A = (2, 3)$, $B = (-1, 4)$, $C = (-3, -2)$, $D = (4, -1)$. Identify their quadrants.
2. Compute the vector from $P = (1, 5)$ to $Q = (6, 2)$.
3. Compute the distance between $A = (-2, 1)$ and $B = (4, 9)$.
4. Find the midpoint of $P = (-3, 7)$ and $Q = (5, -1)$.
5. Compute $\mathbf{u} + \mathbf{v}$ and $2\mathbf{u} - \mathbf{v}$ for $\mathbf{u} = (1, -2, 4)$ and $\mathbf{v} = (3, 0, -5)$.
6. Find the norm of $\mathbf{v} = (2, -3, 6)$.
7. Decide whether $\mathbf{u} = (1, 2)$ and $\mathbf{v} = (4, -2)$ are perpendicular.
8. Write a parametric equation for the line through $P = (0, 3)$ with direction $\mathbf{v} = (2, -1)$.
9. Determine whether $P = (1, 2, 3)$ lies on the plane $x - y + z = 2$.
10. Write the equation of the circle with center $C = (-1, 4)$ and radius 5.

More conceptual exercises

11. Explain why $\mathbb{R}^2$ and $\mathbb{R}^3$ have different dimensions.
12. Give an example of a vector in $\mathbb{R}^5$ and compute its norm.
13. Find a nonzero vector perpendicular to $(5, 2)$.

14. Find a point on the line $(x, y) = (1, 2) + t(3, -4)$ for $t = 2$ and for $t = -1$.
15. Give a geometric interpretation of the equation $x^2 + y^2 + z^2 = 9$.
16. Explain why $x^2 + y^2 \leq 4$ describes a region rather than only a curve.
17. Suppose a data point is represented by $(180, 75, 22)$, where the coordinates mean height in cm, mass in kg, and age in years. What is the dimension of the feature space?
18. Let $P = (1, 1)$ and $Q = (7, 4)$. Find the point one third of the way from P to Q.
19. Find the normal vector to the plane $4x - 2y + 7z = 10$.
20. Compute the angle between $\mathbf{u} = (1, 0)$ and $\mathbf{v} = (1, 1)$.

22 Answers to exercises

1. A: first quadrant, B: second quadrant, C: third quadrant, D: fourth quadrant.
2. $\overrightarrow{PQ} = (6-1, 2-5) = (5, -3)$.
3. $d(A,B) = \sqrt{(4+2)^2 + (9-1)^2} = \sqrt{36+64} = 10$.
4. $M = \left(\frac{-3+5}{2}, \frac{7-1}{2}\right) = (1, 3)$.
5. $\mathbf{u} + \mathbf{v} = (4, -2, -1)$ and $2\mathbf{u} - \mathbf{v} = (-1, -4, 13)$.
6. $\|\mathbf{v}\| = \sqrt{2^2 + (-3)^2 + 6^2} = \sqrt{49} = 7$.
7. $\mathbf{u} \cdot \mathbf{v} = 1 \cdot 4 + 2 \cdot (-2) = 0$, so they are perpendicular.
8. $(x, y) = (0, 3) + t(2, -1)$, or $x = 2t$, $y = 3 - t$.
9. $1 - 2 + 3 = 2$, so yes, P lies on the plane.
10. $(x+1)^2 + (y-4)^2 = 25$.
11. $\mathbb{R}^2$ needs two coordinates; $\mathbb{R}^3$ needs three independent coordinates.
12. Example: $\mathbf{v} = (1, 2, 0, -3, 4)$. Its norm is $\sqrt{1+4+0+9+16} = \sqrt{30}$.
13. One answer is $(2, -5)$, because $(5, 2) \cdot (2, -5) = 10 - 10 = 0$.
14. For $t = 2$: $(7, -6)$. For $t = -1$: $(-2, 6)$.
15. It is a sphere centered at the origin with radius 3.
16. It describes all points whose distance from the origin is at most 2, so it is a disk.
17. The feature space has dimension 3.
18. The point is $(1, 1) + \frac{1}{3}((7, 4) - (1, 1)) = (1, 1) + \frac{1}{3}(6, 3) = (3, 2)$.
19. The normal vector is $(4, -2, 7)$.
20. $\cos\theta = \frac{(1,0)\cdot(1,1)}{\|(1,0)\|\|(1,1)\|} = \frac{1}{\sqrt{2}}$, so $\theta = 45°$.

23 Final perspective

Cartesian space is one of the basic interfaces between mathematics and the quantitative sciences. Once geometry is encoded by coordinates, we can calculate distances, directions, angles, intersections, transformations, and constraints. This is why the same ideas appear in analytic geometry, physics, engineering, graphics, data science, and optimization.

The most important habit is to translate carefully between words, pictures, and formulas:

$$\text{point} \leftrightarrow \text{coordinates}, \qquad \text{displacement} \leftrightarrow \text{vector}, \qquad \text{shape} \leftrightarrow \text{equation}.$$